LAB CSE 4108 — Structured Programming I Lab — Linux & GCC Cheat Sheet

2026-10-31

COMMANDS

`pwd`	current directory
`ls`	list files/folders
`cd`	change directory
`tree`	folder structure as a tree
`cp`	copy files/folders
`mv`	move/rename files/folders
`rm`	delete files
`mkdir`	create a new folder
`touch`	create an empty file
`rmdir`	delete an empty folder
`cat`	print whole file content
`less`	scroll through a large file
`head` / `tail`	first/last lines of a file
`nano` / `vim`	terminal text editors
`chmod`	change file permissions
`chown`	change file owner
`find`	search for files
`grep`	search text inside files
`locate`	fast indexed file search
`man`	manual for a command
`gcc`	compile C code
`gdb`	debug a compiled program
`make`	build multi-file projects
`apt`	install/remove/update packages
`dpkg`	installed-package info
`ps`	list running processes
`top` / `htop`	live CPU/memory usage
`kill`	stop a process by PID
`nohup`	keep running after terminal closes
`jobs` / `fg` / `bg`	manage background jobs
`df -h`	disk space usage
`du -sh`	size of a folder
`free -h`	RAM usage
`ssh`	remote login
`cron`	schedule automatic tasks
`ping`	test connectivity
`curl` / `wget`	download/request from web
`git`	version control

SYMBOLS

`-x`	short flag — e.g. `ls -l`
`--word`	long flag — e.g. `--help`
`>`	redirect output to file, overwrite
`>>`	redirect output to file, append
`<`	take input from file
`|`	pipe — feed output into next command
`&`	run command in background
`&&`	run next command only if previous succeeded
`;`	run commands one after another regardless
`*`	wildcard — matches anything
`~`	home folder
`.`	current folder
`..`	parent folder
`/`	root folder / path separator
`$`	variable value — `echo $PATH`
`$1` `$2`	script arguments
`#`	comment line
`$?`	exit code of previous command (0 = ok)

GCC FLAGS

`-o name`	set output executable name
`-Wall`	show all warnings
`-g`	include debug info (for `gdb`)

সারাংশ: টার্মিনাল নেভিগেশন + ফাইল অপারেশন + gcc/gdb/make + redirection/pipe সাইনগুলো দৈনন্দিন লাগবে। GUI-তে না গিয়ে টার্মিনাল দিয়ে ল্যাবের C ফাইল কম্পাইল-রান প্র্যাকটিস করলে দ্রুত অভ্যস্ত হবে।